

SOFTWARE ENGINEERING

Lab 3: Component Modelling & Architectural Pattern Selection

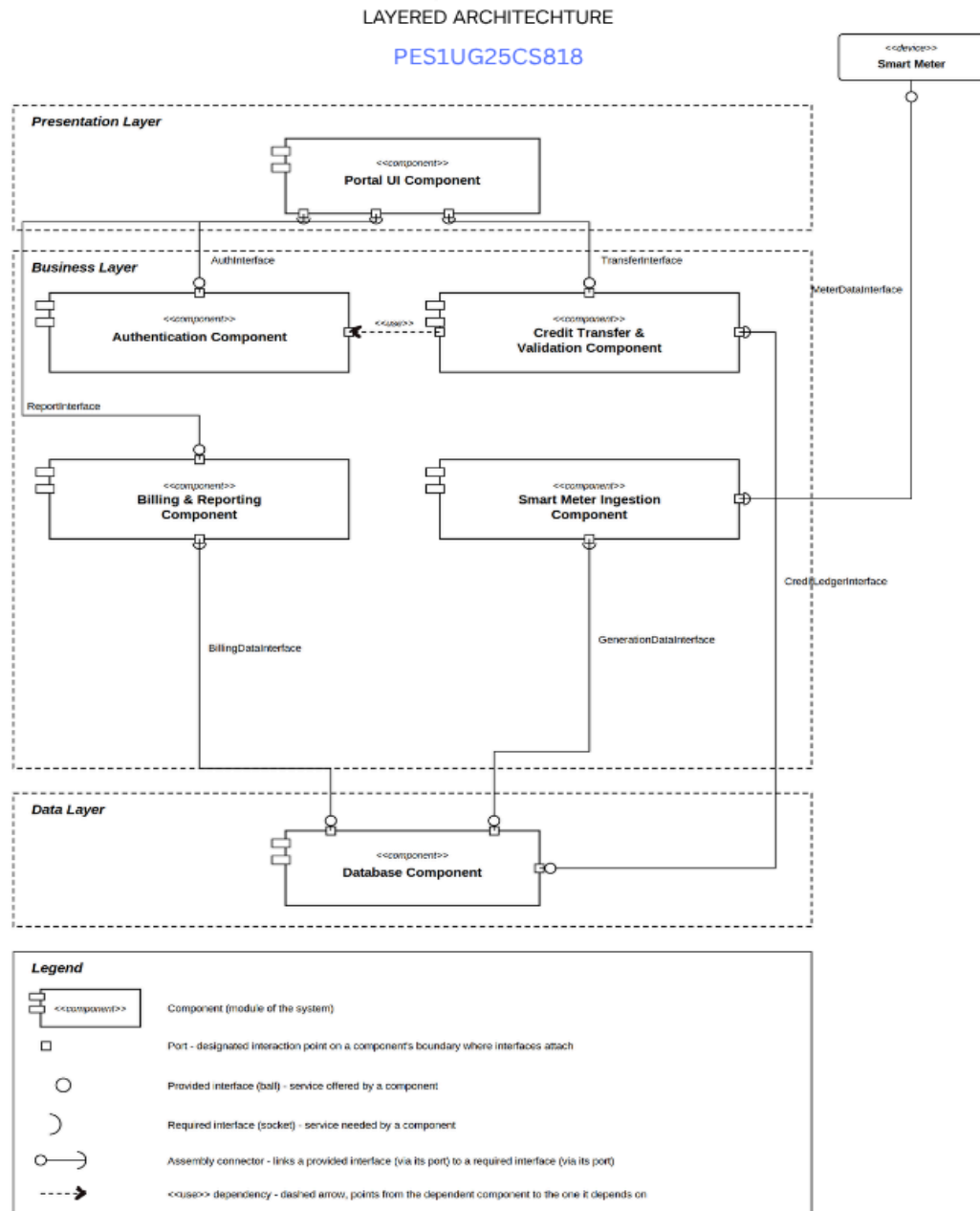
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PROBLEM STATEMENT #63 | Sustainability & Green Tech

Community Solar Credit Allocation Manager



Architecture Selection: Layered Architecture for the Community Solar Credit Allocation Manager System.

Architectural Choice

I organized the system into three layers: a Presentation Layer (Portal UI Component), a Business Layer (Authentication, Credit Transfer & Validation, Billing & Reporting, and Smart Meter Ingestion Components), and a Data Layer (Database Component). Layered Architecture was chosen over Microservices and Client-Server because it maps cleanly onto our requirements at co-op scale without the operational overhead a distributed service mesh would add for a system this size.

Two Reasons

- Requirement fit: our FRs split naturally into three concerns : user interaction (dashboard and transfer requests, FR-002), business rules (validation, approval thresholds, billing calculation FR-001, FR-003, FR-004), and persistent records (credit balances, generation history). Layered Architecture maps directly onto Presentation/Business/Data without forcing artificial service boundaries the way Microservices would.
- Data consistency: prosumers and the Co-op Manager both rely on one accurate, auditable credit ledger (FR-003 billing offsets, NFR-002 encrypted transactions). Funnelling every write through a single Business Layer to one Database Component avoids the cross-service consistency problems that splitting Credit Transfer and Billing into independently-deployed microservices with separate datastores would introduce.

Security Advantage

Centralizing authentication and transfer validation in the Business Layer means every request passes through one chokepoint , the Authentication Component, which Credit Transfer & Validation explicitly depends on via a <<use>> relationship before it ever reaches the Data Layer. This directly satisfies FR-005 (authenticates before granting portal access) and keeps AES-256 encryption (NFR-002) enforced in one place instead of duplicated across independently deployed services, which shrinks the attack surface.

Performance Benefit

Isolating the Data Layer behind the Business Layer lets us tune the Database Component independently , for example, indexing and caching the smart-meter ingestion path (NFR-001: 500 households, 99.9% uptime) without that load interfering with transfer-processing latency. This supports FR-002's requirement that the dashboard reflect updated balances within one minute, since read and write paths stay cleanly separated by layer instead of being tangled across services.